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Examples: Cnidarians (sea anemone, jelly fish) and Ctenophores (comb jellies).

Grade: 2. Bilateria

The eumetazoans other than Radiata, show organ level of organisation and are bilaterally symmetrical and triploblastic. The grade Bilateria includes two taxonomic levels called **Division.**

Division: 1. Protostomia (*Proto*: first; *stomium*: mouth)

Protostomia includes the eumetazoans in which the embryonic blastopore develops into mouth. This division includes three subdivisions namely acoelomata, pseudocoelomata and schizocoelomata.

Division: 2. Deuterostomia (*deuteron*: secondary; *stomium*: mouth)

Eumetazoans in which anus is formed from or near the blastopore and the mouth is formed away from the blastopore. It includes only one subdivision Enterocoelomata. They have a true coelom called enterocoel, formed from the archenteron.

2.3 Non Chordates (Invertebrata)

2.3.1. Phylum: Porifera

(L. *poros*-pore; *ferre*-to bear)

These pore bearing animals are commonly called sponges. They are aquatic, mostly marine, asymmetrical and a few species live in freshwaters. They are primitive, multicellular, sessile animals with cellular level of organisation in which the cells are loosely arranged. They are either radially symmetrical or asymmetrical animals.

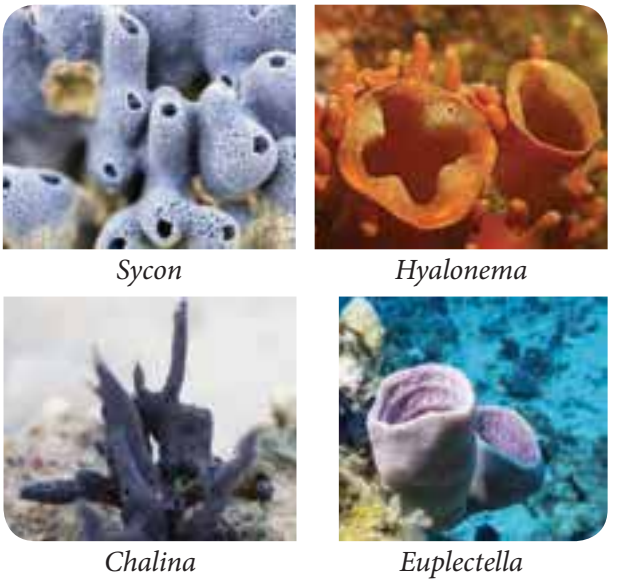


Figure 2.9 Examples of Porifera

They possess a water transport system or **canal system** where water enters through minute pores called **ostia** lining the body wall through which the water enters into a central cavity (**spongocoel)** and goes out through the osculum. This water transport system is helpful in food gathering, circulation, respiration and removal of waste. **Choanocytes** or collar cells are special flagellated cells lining the spongocoel and the canals. The body is supported by a skeleton made up of calcareous and siliceous spicules or spongin or both. Nutrition is holozoic and intracellular. All sponges are hermaphrodites (i.e.) the ova and sperms are produced by the same individual. They also reproduce asexually by fragmentation or gemmule formation and sexually by the formation of gametes. Development is indirect with different types of larval stages such as **parenchymula and amphiblastula**.

Examples: *Sycon* (Scypha), *Spongilla* (fresh water sponge), *Euspongia* (bath sponge) *Euplectella* (Venus flower basket) (Figure 2.9).